



Economics and ecology: Modelling of continuous primary recovery and capture scenarios for recombinant antibody production

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ABSTRACT

With the maturation of antibody production technologies, both economic optimization and ecological aspects have become important. Continuous downstream processing is a way to reduce the environmental footprint and improve process economics. We compared different primary recovery, capture, and fermentation methods for two output-based antibody production scales: 50 kg/year and 1000 kg/year. In addition, a fixed fermentation volume case of 1000 L was analysed in terms of total cost of goods and process mass intensity as a measure of the environmental footprint. In our scenario, a significant amount of water can be saved in downstream processing when single use equipment is utilized. The overall economic and ecological impact is governed by the product titre in our perfusion (1 g/L) and fed-batch (4 g/L). A low titre in fermentation with similar downstream purification leads to higher process mass intensity and cost of goods due to the higher media demand upstream. The economic perspective for continuous integrated biomanufacturing is very attractive, but environmental consequences should not be neglected. Here, we have shown that perfusion has a higher environmental footprint in the form of water consumption compared to fed-batch. As general guidance to improve process economics, we recommend reducing water consumption.

1. Introduction

Process development in the biopharmaceutical industry is not only guided by economic considerations, but also by ecological considerations. Traditionally, the upstream and downstream processing of a biopharmaceutical is batch-based, but in recent years several groups have demonstrated that upstream, as well as downstream, processing can successfully be performed continuously (Hammerschmidt et al., 2016; Shukla et al., 2017; Walther et al., 2015). However, how process economics and ecological factors are affected by a change from batch to continuous manufacturing is unclear. It is also reasonable to look at upstream and downstream processing separately and consider hybrid processes with continuous upstream and batch-based downstream or vice versa. By end-to-end continuous biomanufacturing, the cost of goods (CoG) can be reduced by 40 % for a 10-year scenario (Arnold et al., 2019; Walther et al., 2015).

Although the economic perspective for continuous integrated biomanufacturing is very attractive, the environmental consequences have mostly been neglected. In 2017, approximately 60 % of the market share of all biopharmaceuticals were antibodies and more thorough

investigation of the consequences of continuous biomanufacturing on antibody processes is needed (Grand Review Search, 2017). Platform downstream processes have been established for recombinant antibodies (Biophorum Operations Group, 2014). Such a platform process consists of primary recovery, which is accomplished mainly by centrifugation with a disk stack centrifuge, followed by protein A capture chromatography, a virus inactivation step, at least two additional chromatography steps for intermediate purification and polishing, viral filtration, and an ultrafiltration/diafiltration step (Shukla et al., 2006). Although such processes are termed platform processes, the actual execution, design of primary recovery selection by chromatography, and selection of filters depends on the available facilities, manufacturing scale, product titre, and product characteristics (Farid, 2007; Gottschalk, 2017, 2008; Shukla et al., 2006). The implementation of unit operations that replace stainless steel devices with disposables, as well as continuous process development, create new possibilities for process optimization and further increase the variety in existing facilities. Due to its high cost, the substitution or improvement of protein A chromatography capture is a challenge for the purification of monoclonal antibodies (mAbs). Periodic counter-current chromatography

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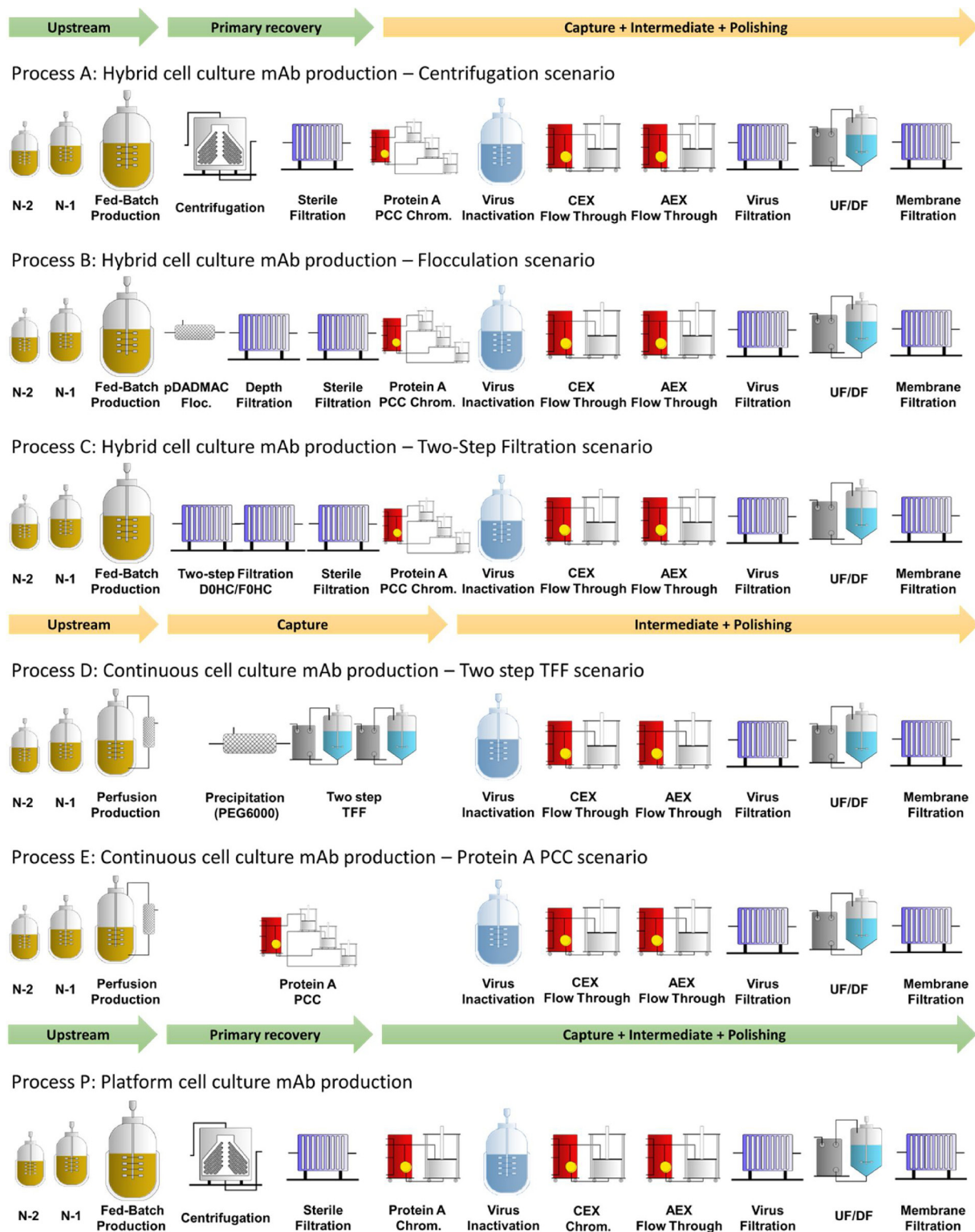


Fig. 1. Process scenarios for antibody production. The yellow arrows above the scenarios represent a continuous product flow, whereas the green arrows represent batch operations. Process A: Batch Upstream followed by conventional centrifugation and filtration for clarification. Process B: Batch Upstream followed by pDADMAC flocculation and two filtration steps. Process C: Batch Upstream followed by D0HC/F0HC filtration and sterile filtration. Process D: Perfusion Upstream followed by polyethylene glycol (PEG) precipitation in a tubular reactor and two diafiltration steps. Process E: Perfusion Upstream followed by a Protein-A PCC. Process F: Platform mAb batch production.

(PCC) is a semi-continuous method to reduce the column volume via higher column utilization and faster process operation. However, the process does become more complex because of additional valves, pumps, and multiple streams (Jungbauer, 2013; Steinebach et al.,

2016). Affinity chromatography can be substituted by alternative capture methods, such as continuous precipitation using a tubular reactor followed by a two-step tangential flow filtration for precipitate harvest. This method allows fully continuous operation with a reduced footprint

without using protein A or columns (Burgstaller et al., 2019; Hammerschmidt et al., 2016).

For batch upstream processes, disk stack centrifugation is the most common unit operation to clarify the harvest from cells (Turner et al., 2018). Single-use substitutes for this unit operation can be cell flocculation combined with depth filtration or a sequence of various depth filters. With the polymer polydiallyldimethylammonium chloride (pDADMAC), the cells can be flocculated and removed by microfiltration. This process requires a smaller filter area than the removal of cells by microfiltration alone. The advantage of pDADMAC is the wide operating range and high yields (Burgstaller et al., 2018; EMD Millipore, 2016; McNerney et al., 2015; Singh and Peck, 2014; Tomic et al., 2014). Thus, it is obvious to compare flocculation-assisted cell removal to centrifugation.

The ecological impact is also a crucial element that needs to be considered in process evaluation. For other industry sectors, the process mass intensity (PMI) has been used to compare the environmental impact of different processes. It is a metric that describes the total resource consumption in mass per mass product (Eq. (1)) (Budzinski et al., 2018).

$$\text{Total PMI} = \frac{\text{total water, raw materials, consumables used in process (kg)}}{\text{active pharmaceutical ingredient (kg)}} \quad (1)$$

Recently, the use of PMI for comparing different processes in biomanufacturing was suggested. In biopharmaceutical manufacturing, the highest contribution to the PMI is water consumption. Reducing the amount of water required has not only economic benefits due to less storage and water for injection (WFI) production effort, but is also environmentally aware (Budzinski et al., 2018; Jimenez-Gonzalez et al., 2011; Madabhushi et al., 2018).

For a meaningful comparison, it is important to define certain production scenarios, either greenfield or with an existing plant. High upfront investment may bias cost comparisons towards processes with disposable equipment, in which the CoG may actually dominate the overall costs. The cost structure will also depend on the phase of development and whether the product is produced for clinical trials or full-scale manufacturing over a long period of time. Nevertheless, the annual manufacturing amount will be crucial for selecting the most economic and environmentally friendly process. We used the computer program BioSolve, which allows custom calculations for different production scenarios of different scales. The models are based on existing data and data from the BioSolve database, which is regularly updated by the developers (Pollard et al., 2016).

In the present study, several batch and continuous manufacturing options for the production of monoclonal antibodies were compared. Possible process parameters and sequences were designed in discussion with our industry partners to operate a 1000 L fermentation broth within 3 days. For primary recovery, centrifugation and flocculation-assisted removal of the cells were compared. Next, capture and polishing steps in bind-elute and flow-through were investigated in batch and continuous operation. Two reference output scales were selected to demonstrate the impact of different scenarios for an orphan substance with an annual demand of 50 kg/year and a blockbuster substance with an annual demand of 1000 kg/year for a greenfield scenario in which a new facility is built (Farid, 2007; Gottschalk, 2017). In addition, the output and cost of the designed facility equipped with a 1000 L cultivation vessel is assessed using a conventional batch platform and the best options for each of the unit operations. We investigated whether the frequently used PMI and CoG positively correlate. If this proves to be true, a process with a small environmental footprint is also economically favourable.

2. Materials and methods

2.1. Process data

The economic evaluation is based on data generated in the EU-funded Horizon 2020 project NextBioPharm DSP (<http://nextbiopharmdsp.eu/>), in which academic institutions cooperate with research and manufacturing companies, having published certain process options and new unit operations (Burgstaller et al., 2019, 2018). All process scenarios were tested in lab-scale and applied in a pilot-scale facility. Based on the results of both scales and the BioSolve database, process parameters were realistically estimated in industrial-scale.

Process definitions of continuous capture via precipitation and pDADMAC-based flocculation scenarios are available in previous publications by Burgstaller et al. (Burgstaller et al., 2019, 2018). Fig. 1 shows all scenarios evaluated in this publication and the unit operations within the sequence. Each scenario consists of a hybrid process with fed-batch fermentation, followed by a continuous downstream train, or is run completely continuous, replacing the production fed-batch fermentation with a perfusion reactor.

For all hybrid scenarios, several primary recovery solutions were compared in a fermentation volume-based scenario of 1000 L, and in two output-based scenarios of 50 kg/year and 1000 kg/year for a direct unit operation comparison. The best performing solution was then applied in a simulated production of the corresponding scale. Capture and polishing were performed continuously and scaled based on the fermentation output to minimize the idle time of each unit operation. The evaluation of the hybrid processes focused on the comparison between different primary recovery solutions; thus, for all scenarios, a protein A capture PCC skid prototype was applied. The perfusion scenarios focused on the comparison between the different capture solutions. Therefore, the stated PCC was compared to continuous precipitation, followed by two-step tangential flow filtration (TFF) (Burgstaller et al., 2019). Subsequently, continuous virus inactivation was performed, followed by flow-through CEX combined with an active carbon filter, pH exchange, flow-through AEX, viral filtration, diafiltration, and a final membrane filtration. The entire continuous purification train, except the two-step filtration in the precipitation scenario (Process D) via Äkta Flux®, was performed using continuously operating prototypes produced by Merck KGaA. Table 1 provides the specific process parameters and modelling inputs. For all scenarios performed in a single-use reactor, the equipment is assumed to be occupied for 90 % of the available time, 80 % for all stainless steel scenarios. For all perfusion scenarios, we assumed a total runtime of 20 days, including 5 days for the start-up phase. We also evaluated a default mAb process of BioSolve as a reference and compared it with our results (Process P).

2.2. Cost data and modelling

Economic modelling requires a considerable amount of process and equipment data. The cost of equipment for each device and the operational costs for all utilized consumables, materials, personnel, and utilities need to be analysed. Therefore, it is important that certain assumptions be realistic to avoid misleading results. The modelling software BioSolve (v7.5) by BioPharm Services was utilized for data that could not be determined otherwise. This software provides a dataset based on yearly cost reports from the industry, which allows the user to minimize the number of errors based on non-comprehensible assumptions. We generated data on the 1000 L fermentation scale, and all relevant process costs could be determined directly from our experiments. All other production scenarios were scaled based on the data generated from 1000 L and complemented with BioSolve data when necessary. Consumables were scaled according to their use in the scenario (e.g., flexware assemblies were used according to the corresponding skid and lifetime, whereas the amount of protein A resin was scaled for the amount of product). For the entire process design we used

Table 1
Scenario parameters.

	Scenario	Capture method	USP titre	DSP yield ^a			Annual output	USP volume
				Total	Prim.	Cap.		
A	Fed-Batch Centrifugation	PCC	4 g/L	81 %	95 %	95 %	50 kg/year 1000 kg/year 81 kg/year	650 L (SU) 2 × 8000 L (SS) 6 × 2000 L (SU)
B	Fed-Batch pDADMAC-Flocculation	PCC	4 g/L	81 %	95 %	95 %	50 kg/year 1000 kg/year 81 kg/year	650 L (SU) 8 × 2000 L (SU) 1000 L (SU)
C	Fed-Batch Two-step Filtration	PCC	4 g/L	81 %	95 %	95 %	50 kg/year 1000 kg/year 81 kg/year	650 L (SU) 8 × 2000 L (SU) 1000 L (SU)
D	Perfusion^b Precipitation + TFF	Prec. + TFF	1 g/L	86 %	–	95 %	50 kg/year 1000 kg/year 266 kg/year	200 L (SU) 4000 L (SS) 1000 L (SU)
E	Perfusion^b PCC	PCC	1 g/L	86 %	–	95 %	50 kg/year 1000 kg/year 266 kg/year	200 L (SU) 4000 L (SS) 1000 L (SU)
P	Platform mAb process	Batch Prot. A	4 g/L	54 %	85 %	90 %	50 kg/year 1000 kg/year 54 kg/year	950 L (SU) 2 × 11,000 L (SS) 1000 L (SU)

^a All yields were determined in 1000 L scale production except for scenario P, in which a BioSolve process was utilized as reference.

^b 1.5 vol exchanges per day.

the default values of the BioSolve “parameter” sheet. This sheet contains construction and manufacturing related parameters. Media and buffers were modelled to be prepared once per batch for fed-batch scenarios and once per production day for all perfusion scenarios. No buffer and media concentration was included in this model. For all 1000 kg/year scenarios we estimated a total personnel amount of 25. For all other scenarios we estimated a total personnel amount of 10. The unit operations were scaled according to the actual scenarios (e.g., primary separations were scaled to wet cell weight and not fermentation volume). In addition to the economic evaluation, modelling with BioSolve enabled analysis of the ecological impact, as well as the PMI, which allows a comparison of the total resource consumption of each scenario. Compared to the PMI of Eq. (1), BioSolve also includes cleaning water in the calculations. For all scenarios the PMI calculation of Eq. (2) is applied.

$$TotalPMI = \frac{processwater(kg) + cleaningwater(kg) + rawmaterials(kg) + consumables(kg)}{product(kg)} \quad (2)$$

The price of 1 L water can vary depending on different factors like purity grade, produced amount of clean water, municipal water quality or production handling. The comparison of purchased water to water produced within the facility needs to be considered as well. All of these uncertainty factors may cause a miscalculation of the water price.

To reduce the risk of this miscalculation, a sensitivity analysis was integrated into the modelling to evaluate the impact of the water price for each scenario. The price per litre WFI was compared within the range of 1 €/L and 4 €/L. These ranges were chosen to display concrete changes of the CoG in case the price per L WFI is 4 times higher. This kind of sensitivity analysis simplifies the understanding of cost dedication within BioSolve. Furthermore, it shows a different way apart from the WFI price calculation, which is classically generated by the software. With the collected data, all scenarios were compared based on the price impact per litre WFI, CoG, PMI, footprint, and cost split between Upstream Processing (USP) and Downstream Processing (DSP).

3. Results and discussion

3.1. Definition of scenarios

In our evaluation, we compared different manufacturing scenarios for a platform antibody process (Fig. 1, Process P) and compared this platform process to different primary recovery and capture alternatives (Fig. 1, Process A–C). These scenarios were then converted to a continuous downstream process. As such, we compared perfusion with fed-batch fermentation, a precipitation-based capture scenario with PCC Protein A capture, and single-use primary recovery methods using flocculation and depth filtration with a conventional centrifuge (Fig. 1, Table 1). The actual process was operated on a 1000 L fermentation scale, with which we then modelled different output-based scenarios of 50 kg/year and 1000 kg/year.

3.2. Floor space requirements

For each unit operation, we measured the required floor space for the 1000 L scale. Each unit operation has dedicated air classification varying from class A to D for production and U for non-classified air requirements. The required floor space for each scale was calculated using the physical dimensions of each device and allowing for free space within them, which was expressed as the total occupancy rate in the facility. For each scenario using single-use equipment, we assumed 50 % occupancy due to mobile accessible devices and fewer intermediate tanks. Thus, 50 % of the available floor space of the production hall is filled with equipment, whereas the rest of the floor space is required for personnel. For all stainless steel fermentations, an occupancy rate of 35 % was assumed for more intermediate tanks and equipment accessibility. Case studies, where integrated continuous biomanufacturing processes were evaluated, already showed design principles, where clean room class D can be applied for antibody purification. Ideally fully continuous unit operations do not require any storage tanks. PCC operations however have an interrupted eluate output. When precipitation is used in a fully closed system, we consider reduced air quality requirements from C to D for this unit operation to demonstrate potential differences (Boedeker and Magnus, 2017). The floor space required for each continuous downstream scenario is shown in Fig. 2A. For each scale, continuous upstream processing with perfusion and continuous capture with PCC requires the smallest footprint.

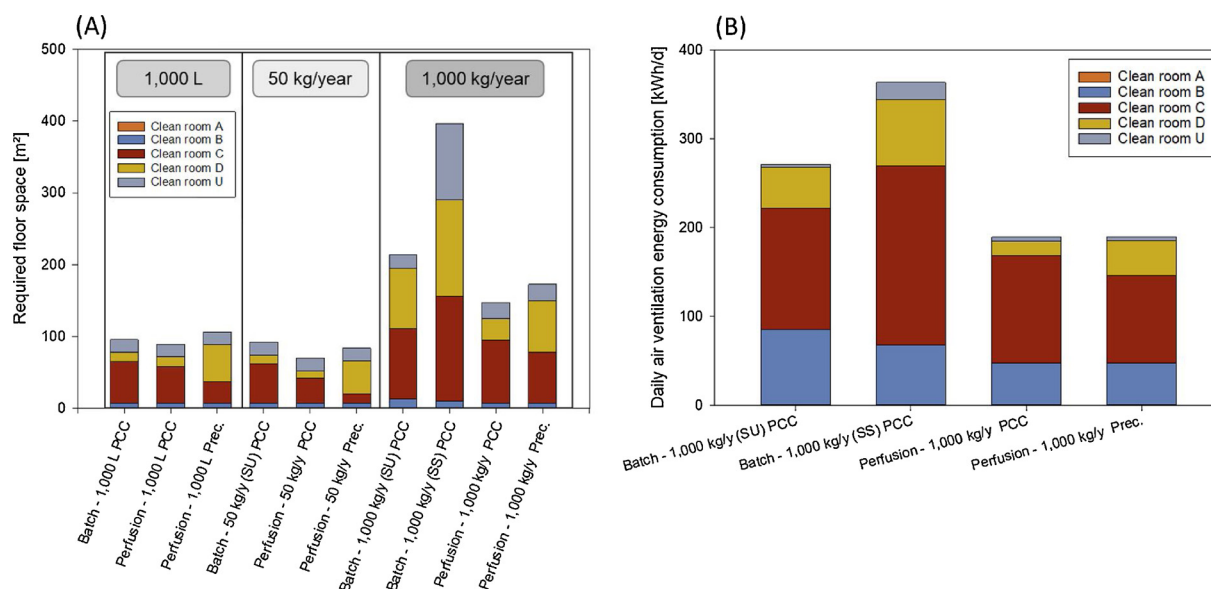


Fig. 2. A: Floor space dedication for each production scenario. All scenarios were calculated with single-use (SU) equipment for fermentation except the 1000 kg/year stainless steel (SS) scenario; B: Energy consumption caused by air ventilation for all 1000 kg/year scenarios.

The required stainless steel buffer and media tanks, as well as the piping and instrumentation for larger bioreactors, cause the difference between the 1000 kg/year stainless steel production and the other scenarios. By switching from a hybrid process of fed-batch upstream and continuous downstream towards fully continuous production, the footprint can be reduced by up to 63 % for stainless steel fed-batch compared to perfusion PCC and 31 % for single-use fed-batch compared to perfusion PCC (Fig. 2A).

According to Annex 6, 2002 WHO report clean room class D can also be classified as class 100,000 and clean room class C as class 10,000 (WHO Expert Committee, 2008). Jaisinghani showed how many air changes per hour are required for each clean room class. Based on these values we determined an average amount of 360/360/75/30/10 air changes per hour for clean room classes A/B/C/D/U (Aircauity guidelines, 2012; Jaisinghani, 2006). Based on the amount of air changes per hour, a room height of 4 m and a 24 h operation per day we calculated the energy consumption for all 1000 kg/year scenarios using a specific fan power of 0.192 Wh/m³ (Pérez-Lombard et al., 2012). Fig. 2B shows that both perfusion scenarios consume less energy caused by air ventilation compared to the fed-batch scenarios.

Nearly all perfusion scenarios have reduced floor space demand compared to all fed-batch scenarios. Due to the water needed for cleaning, the stainless steel fed-batch scenario requires more floor space than the single-use scenario.

Precipitation or PCC have relatively similar floor space requirements, but the air quality requirement is lower for precipitation and will lead to reduced CoG and upfront investment.

3.3. Primary recovery scenarios

We compared three different primary separation setups (Fig. 1, Process A–C) in terms of cost in BioSolve and the ecological footprint on the basis of PMI analysis. The first scenario was a conventional centrifugation step using a disk stack centrifuge (Fig. 1, Process A), which has a high initial capital cost but low running cost. The second scenario was flocculation-assisted depth filtration using pDADMAC (Fig. 1, Process B). The third scenario was conventional depth filtration using a course and fine depth filter for primary separation (Fig. 1, Process C). Capture, intermediate, and polishing were not taken into account because they would not change the evaluation. The volume and product yield after primary recovery for each process option were the same.

Both depth filtration-based separations were run in single-use mode in this scenario, replacing all tubing and filters after each batch, whereas the disk stack centrifuge is used for a lifetime of 10 years. We evaluated different scales of production (expressed in annual production of antibody) for all scenarios and compared the respective cost for each. Depending on the scales and facility, all scenarios had different benefits for certain production scales. Fig. 3 shows the cost difference of both single-use options compared to the disk stack centrifuge for the primary recovery steps. The flocculation-assisted scenario using pDADMAC saves up to 10 % for the CoG compared to the centrifugation for smaller scales of 50 kg/year. For larger scales, however, both single-use solutions increased costs up to 10 %.

The performance of the centrifugation scenario compared to the other scenarios can shift depending on the fermentation volume, titre,

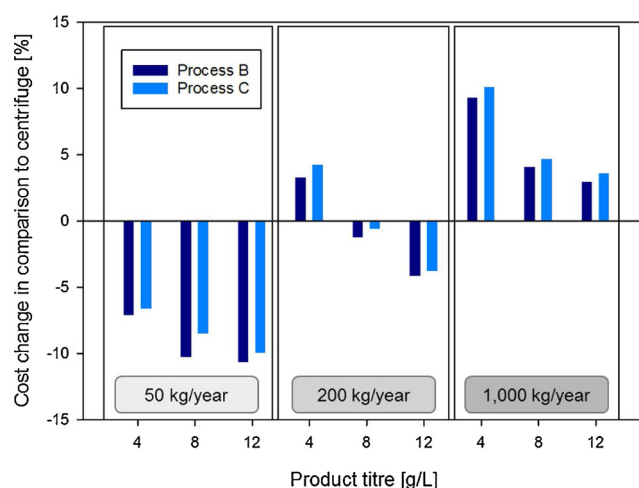


Fig. 3. Relative unit-operation related CoG comparison for the pDADMAC flocculation-assisted microfiltration (Process B) and two-step filtration (Process C) scenarios compared to conventional centrifugation (Process A). The single-use solutions are represented as bars. The conventional centrifugation is implemented as reference and represented by the baseline. Bars above the baseline represent higher CoG of the single-use scenarios compared to the centrifugation and vice versa. The titre varied from 4 g/L to 12 g/L, and the annual output represents an orphan, intermediate, and blockbuster antibody based on the output.

and annual product demand. Various scenarios with different titres and annual product output were compared to the disk stack centrifuge in terms of cost (Fig. 3). With lower annual output (50 kg/year), the improvement in product titre had a small impact on the CoG. The single-use options are more favourable than a disk stack centrifuge for smaller operations (i.e., 50 kg/year). A simulation was added for a production scale of 200 kg/year. Walther and Godawat (Walther et al., 2015) also stated that this is a common manufacturing scale. At low titre, the centrifugation (Process A) has better economic performance compared to single-use solutions (Process B and C). Higher process volumes must be handled at lower titres, increasing the required filter area. The flocculation-assisted depth filtration with pDADMAC consistently outperformed the conventional two-step filtration on all scales and with all titres. The capital costs for the disk stack centrifuge are included in the model by distributing the cost over the number of batches during the life of the centrifuge, which is typically assumed with a 10-year depreciation period. This means that, with higher production per year, the contribution of the capital costs reduces progressively.

The PMIs for the 50 kg and 1000 kg annual production were compared. For both scales, the water consumption for centrifugation was much higher compared to the single-use solutions (Fig. 4). We hypothesized that PMI correlates with the CoG. A correlation between the PMI and CoG was seen for smaller scales (compare Figs. 3 and 4A). The PMI of the primary recovery by filtration and CoG were lower compared to centrifugation. However, the PMI of the centrifugation was the highest but the CoG the lowest with the 1000 kg scale (compare Figs. 3 and 4B). This finding can be explained by the high consumable cost of both single-use solutions. On the large-scale, the centrifugation becomes more efficient and less water is consumed for cleaning the equipment. Total filter costs outweigh the costs of water consumption in centrifugation. In this case, a reduced PMI can be achieved by filtration, but the disadvantage of large expenses outweighs the benefit of the PMI.

3.4. Switching from conventional batch DSP to continuous

Batch platform antibody production was used as a reference (Fig. 1, Process P) and compared to different versions of batch USP and batch primary recovery followed by continuous DSP (Fig. 1, Process A and B). Process C (Fig. 1) was not considered because we have shown that the primary recovery was not better than with Process B. In Fig. 5, the CoG of continuous DSP is compared to the reference process. For each scale, continuous DSP had reduced CoG. Especially for smaller scales and higher titres (12 g/L), the potential cost reduction of 57 % can be achieved by applying the flocculation-assisted filtration. The product yield of the continuous DSP was 81 %, compared to 54 % for the platform reference process from BioSolve. This is a very conservative calculation because the state of the art antibody process may also have

an overall product yield of more than 70 % (Walther et al., 2015). The difference in cost is reduced for higher scales in the course of the contribution of consumable costs. Though only SU bioreactors are used for the continuous scenarios, the platform process for larger scales was operated by stainless steel fermentation. We also took pre-packed columns into consideration, which were compared to acrylic columns for the conventional process. The change from single-use to stainless steel for the platform process can reduce costs for 1000 kg/year scale processes, but the PMI is increased 53 % when comparing 50 kg/year and 1000 kg/year.

3.5. Economics of alternative continuous capture steps

To quantify the influence of future non-chromatographic unit operations, such as precipitation, we used the data generated in our Horizon 2020 project (Burgstaller et al., 2019). Polyethylene glycol (PEG) precipitation was applied as a capture step and is suitable for coupling to an industrial perfusion reactor. We compared this non-chromatographic unit operation (Fig. 1, Process D) with PCC (Fig. 1, Process E) in a completely end-to-end continuous process on different scales: 50 kg/year, 1000 kg/year, and a 1000 L fermentation. The CoG and process water consumption were calculated. For the PCC scenario (Process E), the capture output volume was based on the eluate, whereas for precipitation, a 13-fold concentration was implemented based on lab-scale data. For our model, we assumed that a titre of 1 g/L is reached in the perfusion culture. This assumption is based on the perfusion culture of our Horizon 2020 project, which was run at pilot-scale. The parameters for economic evaluation and process water consumption were obtained from this end-to-end pilot-scale experiments. The CoG and PMI of both scenarios (Process D and E) at different scales are shown in Fig. 6A. For larger scales, the PMI and CoG were higher in the precipitation scenarios than the PCC scenarios. For the 1000 L scale, in which 266 kg/year can be produced, and the 1000 kg/year scale, the increased CoG can be correlated with the PMI. The volume reduction is lower for the precipitation than the chromatographic process (Process E). To keep the residence time constant for the subsequent chromatographic flow-through steps, we had to increase the size of the AEX and CEX columns to process the material at the same time. However, the volume stays constant in precipitation regardless of titre. Therefore, this method has advantages for high titre feedstocks. Budzinski et al. (Budzinski et al., 2018) stated that the PMI increases with the number of chromatography unit operations in a process. Based on this statement we can indicate that increased column volumes with standard bind-elute operation methods lead to a higher water consumption. Increasing the binding capacity of the column will lead to less column volumes and less buffer consumption. For protein A capture we tested MabSelect SuRe, with a dynamic binding capacity of 44 g_{mAb}/L_{Resin} and Amsphere A3, which had a dynamic binding capacity of 65 g_{mAb}/L_{Resin}. The PMI

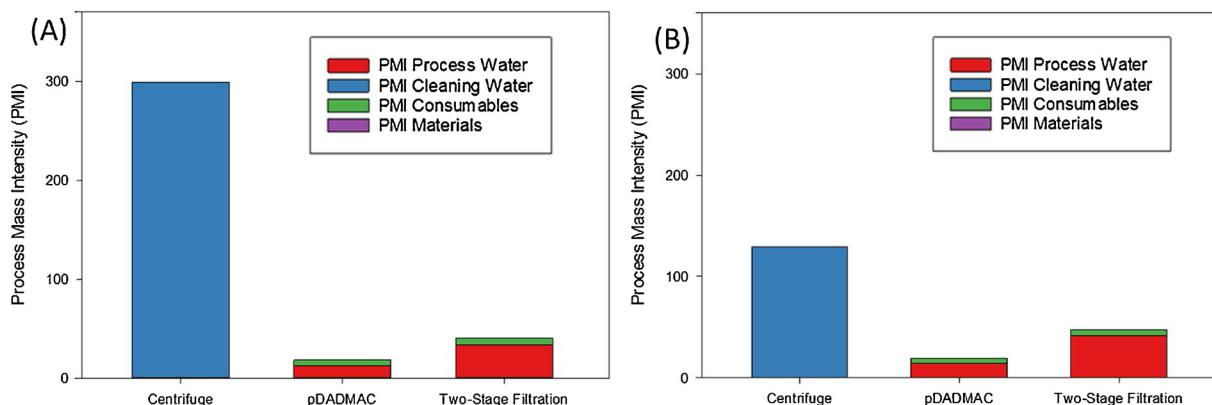


Fig. 4. PMI of primary recovery scenarios. A: 50 kg/year scale; B: 1000 kg/year scale. This figure considers the PMI of the specific primary recovery devices without taking other unit operations into account.

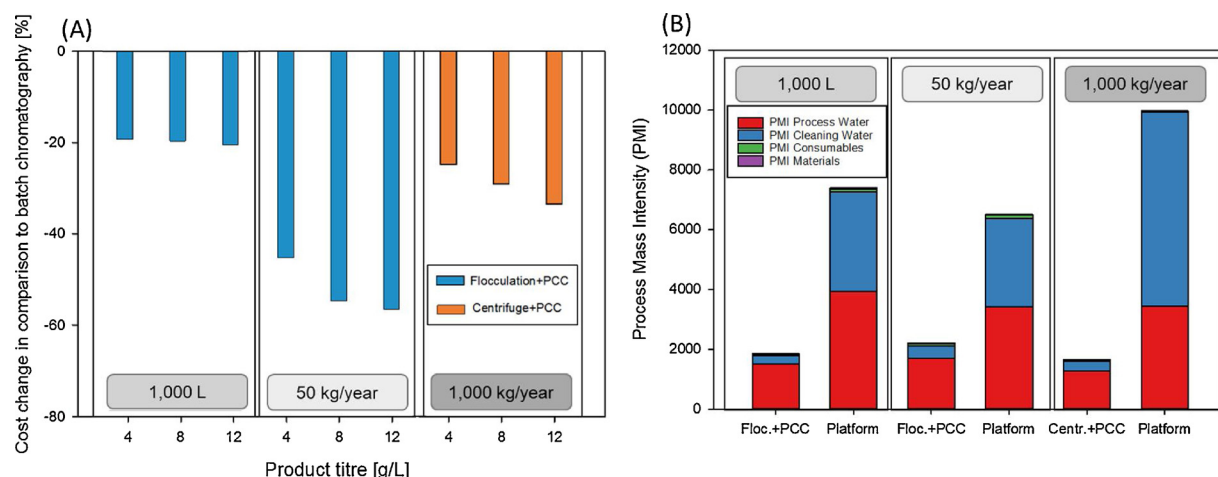


Fig. 5. A: Relative CoG evaluation of the entire production train. Continuous downstream prototype containing Protein A PCC (Process B for 1000 L and 50 kg/year, Process A for 1000 kg/year) was compared to a conventional antibody process using batch protein A chromatography (Process P). The continuous operations are represented as bars. The conventional antibody process is implemented as reference and represented by the baseline. Bars above the baseline represent higher CoG of continuous downstream compared to Process P and vice versa B: PMI analysis of the platform process and continuous purification. This figure considers the PMI of the entire production train.

of MabSelect Sure was thereby higher than the PMI of Amsphere A3. This impact however is reduced for flow-through chromatography steps, where less column volumes for regeneration are required until the column can be reloaded.

For a yearly production of 50 kg, the lower volume reduction and larger polishing columns when using precipitation as a capture step are compensated for by the lower cost of unit operation. Protein A costs dominate the CoG for the total process for the 50 kg/year scale. For each scale, the PMI was significantly higher for the precipitation process than for the PCC (Fig. 6B). For higher scales, the CoG were higher for precipitation than the PCC. In addition, the precipitation performance varies based on the price of the WFI. These numbers do not take into account a potential difference in performance of the polishing unit when switching to alternative technologies, as there is no data available on our process in regards to polishing precipitated material. Therefore, these numbers have to be considered with caution and will need to be partially revised once more data is available in the literature for polishing of precipitated mAb material.

For the 1000 L scale, higher WFI prices had greater CoG differences, and the gap of the 1000 kg/year scale was reduced for higher WFI

prices. In this scenario, a relative comparison of the CoG revealed a price difference of approximately 30 %, whereas the PMI was increased only 15 % for the precipitation. This means that, even though the absolute CoG for the precipitation is higher than for the PCC, the relative difference of both scenarios is reduced. For the 50 kg/year scale, the PMI benefit of the PCC scenario cannot achieve cost savings compared to precipitation because of the impact of the cost of the protein A column.

3.6. Hybrid and perfusion scenarios

We compared a hybrid process to fed-batch fermentation with a product titre of 4 g/L, followed by the best performing primary recovery and PCC for capture at each scale (Process A or Process B) with perfusion fermentation and a product titre of 1 g/L followed by PCC capture (Process E). Primary recovery was not required because the effluent of the perfusion culture could be processed directly by PCC. For the output-based scenarios, however, the fed-batch scenarios had reduced CoG. Fig. 7A shows the CoG comparison of a fully continuous process (Process E) to a hybrid process (Process A or B) and, for the 50

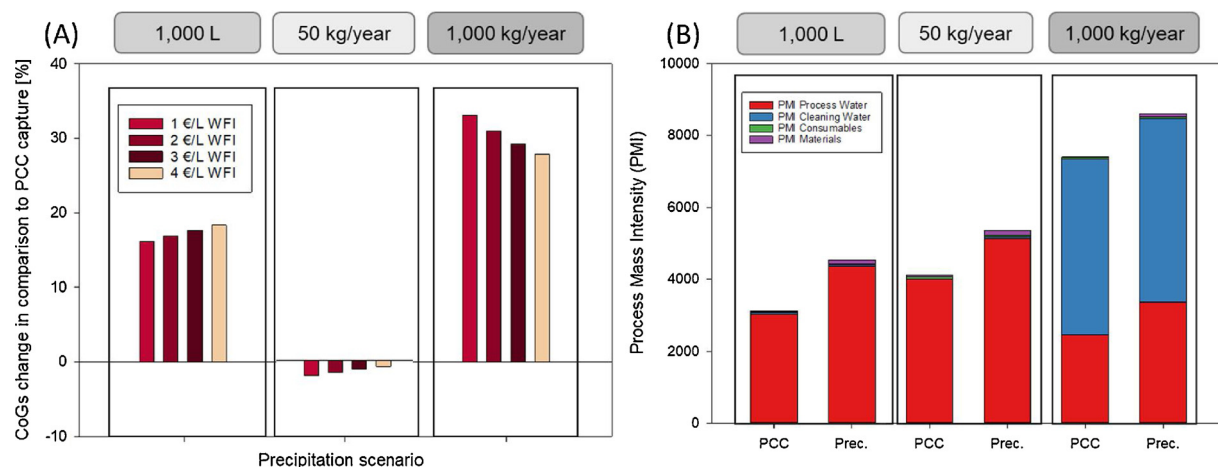


Fig. 6. A: CoG evaluation comparing the Perfusion Precipitation scenario (Process D) and Perfusion PCC scenario (Process E). Process D is represented with bars. Process B is implemented as reference and represented by the baseline. Different water prices were considered for both scenarios. Bars above the baseline represent higher CoG of Process D compared to Process E and vice versa B: PMI analysis of the Perfusion PCC and Perfusion Precipitation scenario. This figure considers the PMI of the entire production train.

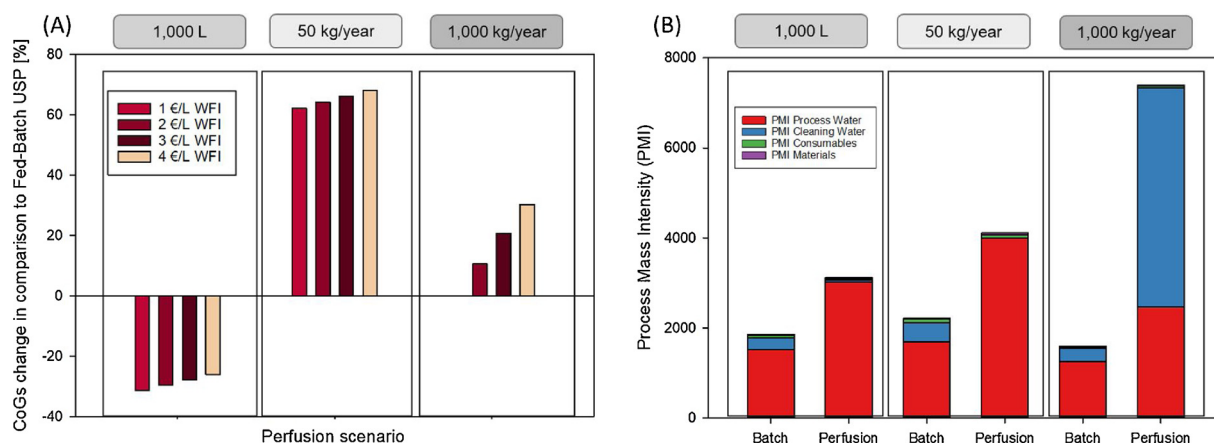


Fig. 7. A: CoG evaluation of the perfusion PCC scenarios (Process E) compared to the fed-batch PCC scenarios. Process E is represented with bars. The best performing fed-batch scenario (Process B for 50 kg/year and 1000 L, Process A for 1000 kg/year) is taken as reference and represented by the baseline. Different water prices were considered for both scenarios. Bars above the baseline represent higher CoG of Process E compared to fed-batch operation and vice versa. B: PMI analysis of the batch PCC and perfusion PCC scenarios. This figure considers the PMI of the entire production train.

kg/year scale, the fully continuous process costs are up to 68 % higher than fed-batch. For a 1000 kg/year scale, the performance of the perfusion scenario depends on the price per litre WFI. For this scale, the WFI price has a significant impact on the CoG. The reason for the sensitivity of perfusion-based processes on the cost of process water is the higher water demand of such processes (Fig. 7B). The relatively low titre of 1 g/L in perfusion processes compared to the 4 g/L for fed-batch leads to an increased demand for media to achieve the same product output as fed-batch and, therefore, higher water consumption. For the 1000 kg/year scale, we replaced all single-use storage bags in the models, which were larger than 2000 L with stainless steel vessels. The perfusion scenario on the 1000 kg/year scale had a significant increase in PMI due to cleaning the stainless steel vessels. By achieving a perfusion titre of 2 g/L, we calculated a reduced CoG for the perfusion on the 1000 kg/year scale of up to 24 % compared to a titre of 1 g/L and reduced PMI of 34 %.

Considering both the overall process water consumption and the CoG of these scenarios, batch processes have an advantage over perfusion processes.

Previous comparisons (Process A–C; Process A, B, P; Process D, E) showed potential water and cost savings for switching from batch-wise operation to continuous operation and from single-use to stainless steel devices. However, continuous upstream operations resulting in lower titres compared to fed-batch have a substantial impact on the economics of fully continuous antibody production (Hammerschmidt et al., 2016).

3.7. Water related process parameters

The PMI displays the total water consumption of a process. However previous results have shown that not only the direct contribution of water is responsible for different CoG. Each unit operation with a relative high amount of consumed water have some high impact cost factors, which are connected to a higher water consumption. For USP a higher PMI is connected to higher media costs. Larger amounts of high cost cell culture media lead to higher CoG. For DSP especially chromatography operations have a strong impact on an increased PMI (see 3.5). Buffer ingredients are hereby the driving costs linked with the water consumption. Throughout the entire process, media and buffers need to be prepared and stored in bags or vessels. The equipment sizing is linked to the amount of required water. Bigger vessels require more floor space and have increased capital costs as well. Also other auxiliary costs such as piping and instrumentation are increased with a higher water consumption. Acquiring all of this information separately increases the quantity of assumptions for the economic evaluation. The

risk of miscalculation caused by assumptions in a wrong dimension increases. Using the PMI as a key metric simplifies the overall amount of assumptions if the cost contribution of single-use consumables is taken into account.

4. Conclusion

From our economic evaluation and analysis of PMI, we conclude that Process A is the best for the large-scale scenario with 1000 kg/year and Process B for the small-scale scenario with 50 kg/year. Switching from fed-batch USP with PCC capture to perfusion with PCC capture reduces the floor space requirements in all of the evaluated scenarios. The fermentation volume-based 1000 L scale is not comparable to the other scales due to different annual product outputs for fed-batch and perfusion USP (81 kg/year and 266 kg/year, respectively). However, perfusion has higher equipment utilization for the DSP equipment, which is one of the biggest benefits of continuous manufacturing compared to batch operation. This leads to improved CoG despite a higher PMI. The hypothesis that the CoG correlates with the PMI is true in all scenarios except when the consumable costs outweigh the water costs. Our evaluation confirms that the PMI can be used to determine potential cost savings in terms of storage floor space, tanks, and labour. Implementation of the PMI can also lead to more sustainable production regarding buffer recycling, which is currently still a burden for the biopharmaceutical industry (Jungbauer and Walch, 2015).

An overall judgement regarding the best performing unit operations and process trains cannot be reached. It is very important to take the annual scale into consideration. The most striking example is primary recovery at small-scale, for which the flocculation-assisted filtration performs better than centrifugation, and at large-scale is the opposite. The economic perspective for continuous integrated biomanufacturing is very attractive, but environmental consequences should not be neglected. Here, we have shown that perfusion has a higher environmental footprint in the form of water consumption compared to fed-batch. As general guidance to improve process economics, we recommend reducing water consumption.

Author contributions

Cataldo A.L: Conducted economic modelling and drafted the manuscript

Burgstaller D: Experiments for continuous antibody capture, basis for parameter estimation of modelling

Hribar G: Provided data for modelling of industrial scale and helped to define the scenario for manufacturing

Jungbauer A: Designed the research project and co-authored the manuscript, overlooked the research

Satzer P: Supervised the project and deliverables of economic modelling

Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

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References

- Aircuity guidelines, 2012. Laboratory Ventilation ACH Rates Standards and Guidelines.
- Arnold, L., Lee, K., Rucker-Pezzini, J., Lee, J.H., 2019. Implementation of fully integrated continuous antibody processing: effects on productivity and COGm. *Biotechnol. J.* 14, 1800061. <https://doi.org/10.1002/biot.201800061>.
- Biophorum Operations Group, 2014. Continued Process Verification: An Industry Position Paper With Example Plan.
- Boedeker, B., Magnus, J., 2017. Opportunities and Limitations of Continuous Processing and Use of Disposables.
- Budzinski, K., Blewis, M., Dahlin, P., D'Aquila, D., Esparza, J., Gavin, J., Ho, S.V., Hutchens, C., Kahn, D., Koenig, S.G., Kottmeier, R., Millard, J., Snyder, M., Stanard, B., Sun, L., 2018. Introduction of a process mass intensity metric for biologics. *New Bioeth.* <https://doi.org/10.1016/j.nbt.2018.07.005>.
- Burgstaller, D., Jungbauer, A., Satzer, P., 2019. Continuous integrated antibody precipitation with two-stage tangential flow microfiltration enables constant mass flow. *Biotechnol. Bioeng.* <https://doi.org/10.1002/bit.26922>.
- Burgstaller, D., Krepper, W., Haas, J., Maszulin, M., Mohoric, J., Pajnic, K., Jungbauer, A., Satzer, P., 2018. Continuous cell flocculation for recombinant antibody harvesting. *J. Chem. Technol. Biotechnol.* 93, 1881–1890. <https://doi.org/10.1002/jctb.5500>.
- EMD Millipore, 2016. Clarification Portfolio Guide.
- Farid, S.S., 2007. Process economics of industrial monoclonal antibody manufacture. *J. Chromatogr. B Analyt. Technol. Biomed. Life Sci.* 848, 8–18. <https://doi.org/10.1016/j.jchromb.2006.07.037>.
- Gottschalk, U., 2017. Process Scale Purification of Antibodies. John Wiley & Sons.
- Gottschalk, U., 2008. Bioprocess separation in antibody manufacturing: the good, the bad and the ugly. *Biotechnol. Prog.* 24, 496–503. <https://doi.org/10.1021/bp070452g>.
- Grand Review Search, 2017. Biopharmaceuticals Contract Manufacturing Market Report. pp. 2018–2025.
- Hammerschmidt, N., Hobiger, S., Jungbauer, A., 2016. Continuous polyethylene glycol precipitation of recombinant antibodies: sequential precipitation and resolubilization. *Process Biochem.* 51, 325–332. <https://doi.org/10.1016/j.procbio.2015.11.032>.
- Jaishingani, R., 2006. Energy efficient low operating cost cleanroom airflow design. IEST's ESTECH 2003 Conference Phoenix.
- Jimenez-Gonzalez, C., Ponder, C.S., Broxterman, Q.B., Manley, J.B., 2011. Using the Right Green Yardstick: Why Process Mass Intensity Is Used in the Pharmaceutical Industry To Drive More Sustainable Processes [WWW Document]. <https://doi.org/10.1021/op200097d>.
- Jungbauer, A., 2013. Continuous downstream processing of biopharmaceuticals. *Trends Biotechnol.* 31, 479–492. <https://doi.org/10.1016/j.tibtech.2013.05.011>.
- Jungbauer, A., Walch, N., 2015. Buffer recycling in downstream processing of biologics. *Curr. Opin. Chem. Eng.* 10, 1–7. <https://doi.org/10.1016/j.coche.2015.06.001>.
- Madabhushi, S.R., Gavin, J., Xu, S., Cutler, C., Chmielowski, R., Rayfield, W., Tugcu, N., Chen, H., 2018. Quantitative assessment of environmental impact of biologics manufacturing using process mass intensity analysis. *Biotechnol. Prog.* <https://doi.org/10.1002/btpr.2702>. 0.
- McNerney, T., Thomas, A., Senczuk, A., Petty, K., Zhao, X., Piper, R., Carvalho, J., Hammond, M., Sawant, S., Bussiere, J., 2015. PDADMAC flocculation of Chinese hamster ovary cells: enabling a centrifuge-less harvest process for monoclonal antibodies. *mAbs* 7, 413–428. <https://doi.org/10.1080/19420862.2015.1007824>.
- Pérez-Lombard, L., Ortiz, J., Maestre, I.R., Coronel, J.F., 2012. Constructing HVAC energy efficiency indicators. *Energy Build.* 47, 619–629. <https://doi.org/10.1016/j.enbuild.2011.12.039>.
- Pollard, D., Brower, M., Abe, Y., Lopes, A.G., Sinclair, A., 2016. Standardized economic cost modeling for next-generation MAb production. *Bioprocess Int.*
- Shukla, A.A., Etzel, M.R., Gadam, S., 2006. Process Scale Bioseparations for the Biopharmaceutical Industry.
- Shukla, A.A., Wolfe, L.S., Mostafa, S.S., Norman, C., 2017. Evolving trends in mAb production processes. *Bioeng. Transl. Med.* 2, 58–69. <https://doi.org/10.1002/btm2.10061>.
- Singh, N., Peck, M., 2014. pDADMAC Flocculant Reagent for Use With Clarisolve® Depth Filters.
- Steinebach, F., Müller-Späh, T., Morbidelli, M., 2016. Continuous counter-current chromatography for capture and polishing steps in biopharmaceutical production. *Biotechnol. J.* 11, 1126–1141. <https://doi.org/10.1002/biot.201500354>.
- Tomic, S., Besnard, L., Fürst, B., Reithmeier, R., Wichmann, R., Schelling, P., Hakemeyer, C., 2014. Complete clarification solution for processing high density cell culture harvests. *Sep. Purif. Technol.* 141. <https://doi.org/10.1016/j.seppur.2014.12.002>.
- Turner, R., Joseph, A., Titchener-Hooker, N., Bender, J., 2018. Manufacturing of proteins and antibodies: chapter downstream processing technologies. *Adv. Biochem. Eng. Biotechnol.* 165, 95–114. <https://doi.org/10.1007/10.2016.54>.
- Walther, J., Godawat, R., Hwang, C., Abe, Y., Sinclair, A., Konstantinov, K., 2015. The business impact of an integrated continuous biomanufacturing platform for recombinant protein production. *J. Biotechnol.* 213, 3–12.
- WHO Expert Committee, 2008. WHO Expert Committee on Specifications for Pharmaceutical Preparations. WHO.